

Fate and Transport: The Atmosphere (sections 4.5 & 4.6)

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Based on text from: Chemical Fate and Transport in the Environment

Mixing Height

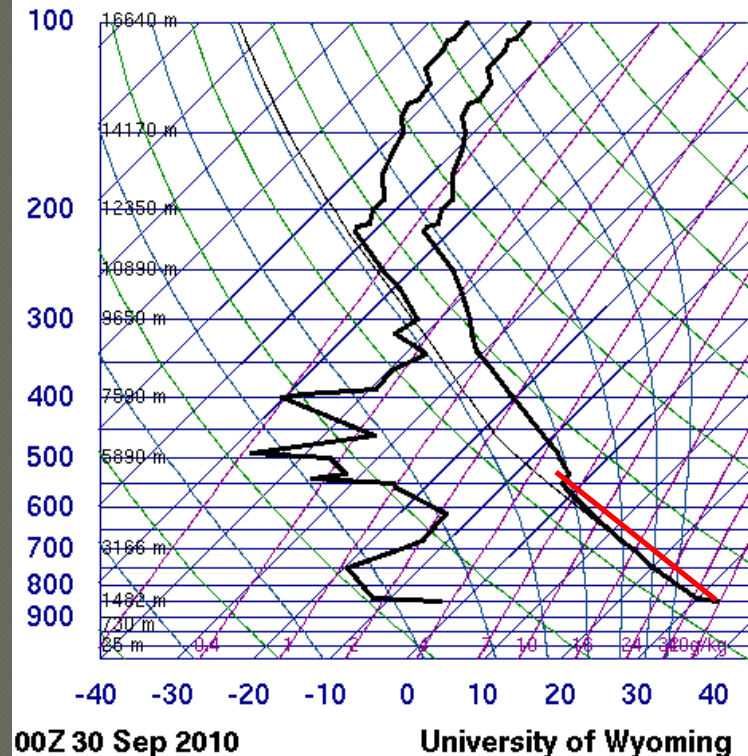
- The Holzworth method

- Provides twice-per-day (morning and afternoon) mixing heights based on calculations using routine NWS upper-air data. The morning mixing height is calculated as the height above ground at which the dry adiabatic extension of the morning minimum surface temperature plus 5 °C intersects the vertical temperature profile observed at 1200 Greenwich Mean Time (GMT).
- Source: http://www.webmet.com/met_monitoring/651.html

- The wet adiabat is used when the dewpoint depression is equal to or less than 5 degrees (Dewpoint depression is $T - T_d$). – John Darnley NOAA

Mixing Height? (surface temp = ~~8~~°C)

72489 REV Reno



SLAT 39.56
SLON -119.80
SELV 1516.
SHOW -9999
LIFT 7.43
LFTV 7.07
SWET -9999
KINX -9999
CTOT -9999
VTOT -9999
TOTL -9999
CAPE 0.00
CAPV 0.00
CINS 0.00
CINV 0.00
EQLV -9999
EGTV -9999
LFCT -9999
LFCV -9999
BRCH 0.00
BRCV 0.00
LCLT 253.3
LCLP 446.7
MLTH 318.9
MLMR 1.86
THCK 5865.
PWAT 6.08

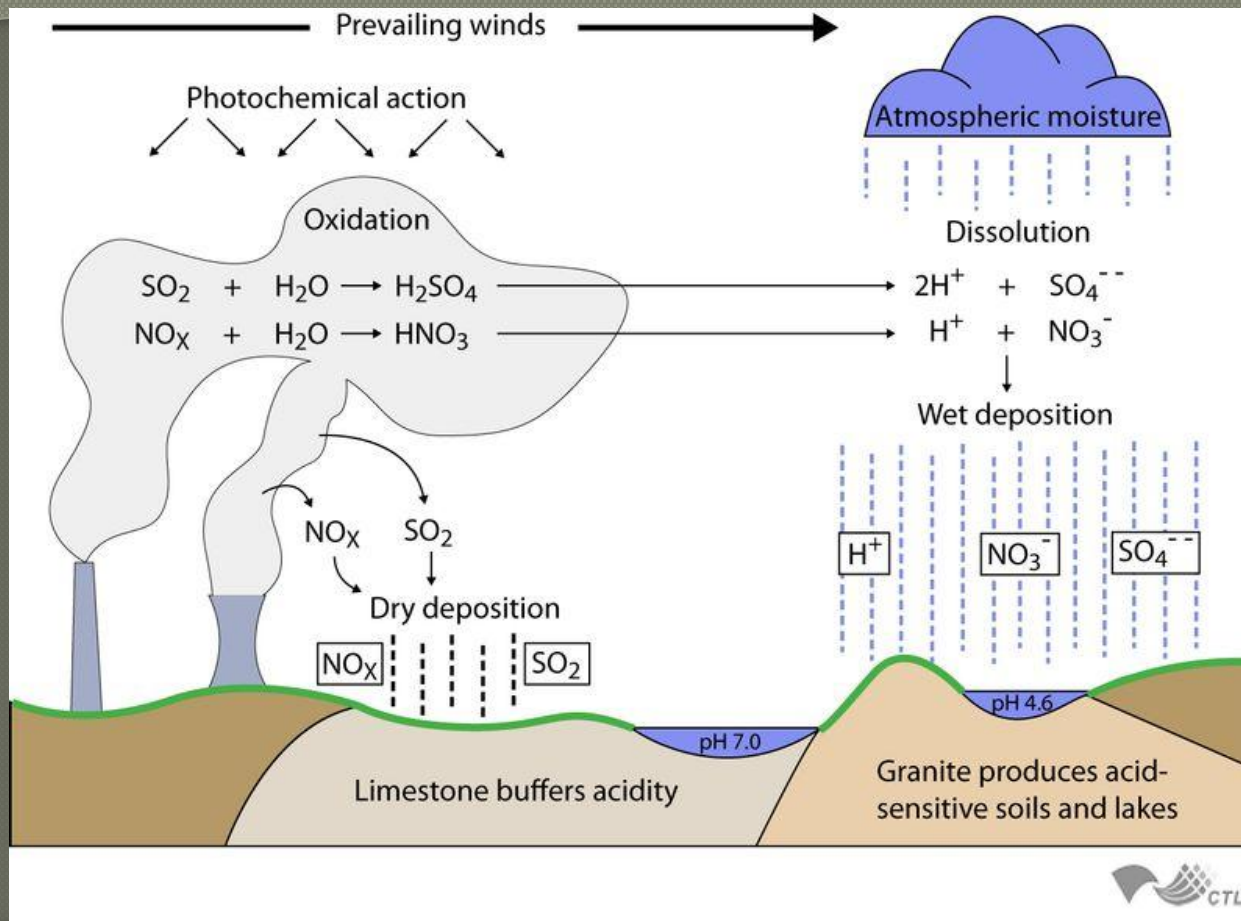
Text from Figure 4-9:
“Draw a line
**originating at the
surface temperature**
with a slope equal to
the applicable
adiabatic lapse rate;
the intersection of the
line with the actual
temperature profile
approximates the
mixing height.”

Source: <http://www.patarnott.com/atms411/notes.htm>

Objectives

● Understand & Discuss:

- Dry deposition (3 key mechanisms)
- Wet deposition
- Characteristics of ozone
- O_3 - NO_x -hydrocarbon cycle
- Acid deposition
- Ozone hole

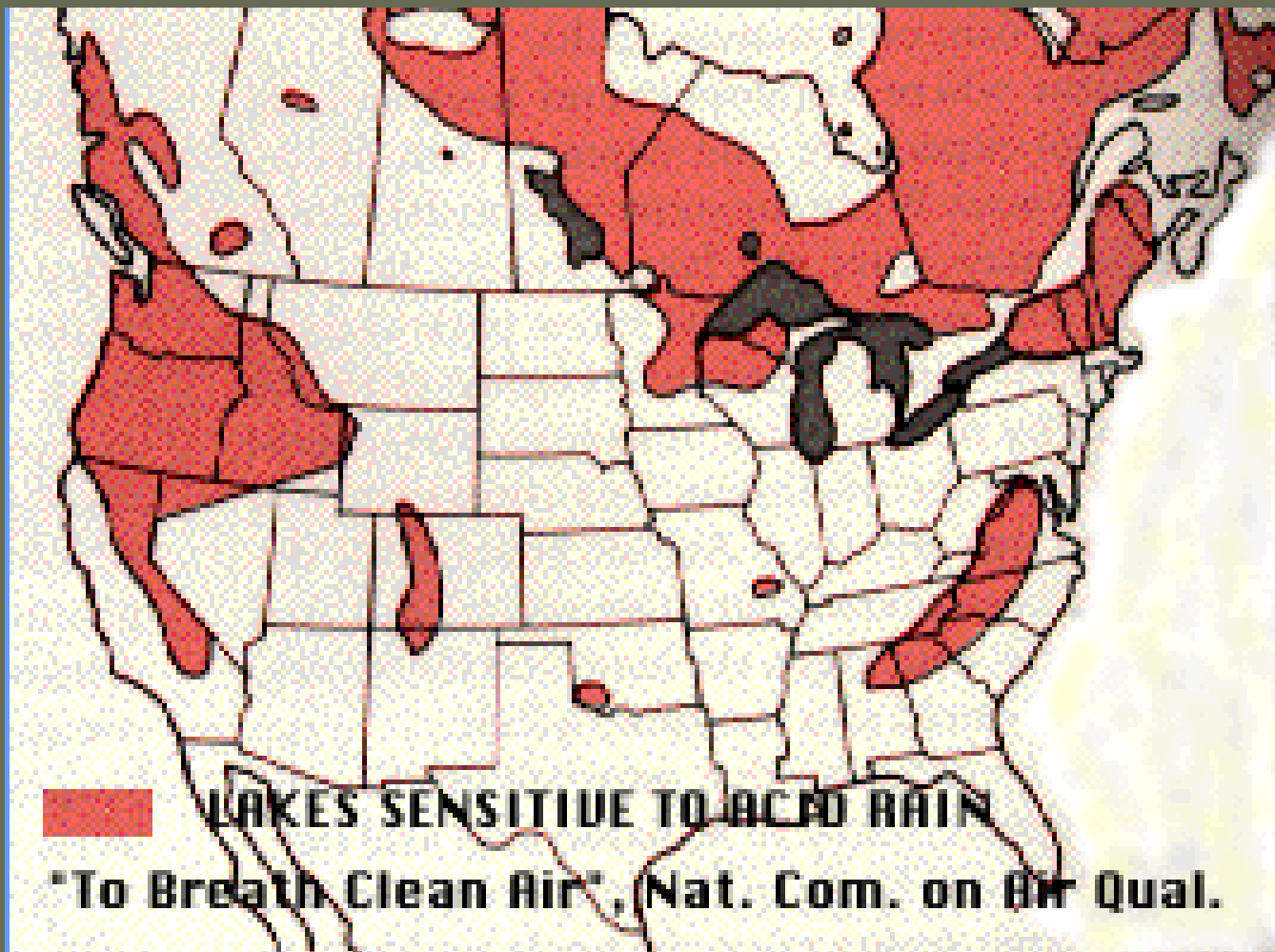


Dry and Wet Deposition of Air Pollutants

Source: "Dry and Wet Deposition of Air Pollutants" from Environmental Health. Available at: <http://ocw.jhsph.edu>. Copyright © Johns Hopkins Bloomberg School of Public Health. Creative Commons BY-NC-SA. Adapted by CTLT from Kemp, D. D.

Location of Lakes Sensitive to Acid Rain

source: <http://www.elmhurst.edu/~chm/vchembook/194acidraineffects.html>



Dry Deposition

● Definition

- Any physical process that does not involve precipitation

● 3 key dry deposition mechanisms:

- Gravitational Settling
- Impaction
- Absorption



An atmospheric deposition sampler placed on the roof of a building.

Source:

<http://www.sccwrp.org/ResearchAreas/Contaminants/AtmosphericDeposition/TraceMetalsDeposition.aspx>

Gravitational Settling

- Significant removal mechanism for particles $> 1\mu\text{m}$
- For particles where $r < \sim 100\mu\text{m}$, grav. settling rate can be estimated using Stoke's Law
 - For air, reduces to convenient form:
 - w_f (settling velocity) = $10^{-2}r^2(\text{cm/sec})$
 - r is the particle radius in μm
 - assume $\Delta\rho = 1 \text{ g/cm}^3$ i.e. difference btwn particle and fluid density
- For particles $r < \sim 1\mu\text{m}$, gravitation settling may be neglected
 - Small particles may be incorporated into larger particles via collision and coagulation; incorporated into raindrops, hailstones, etc; and/or be removed via impaction or diffusive transport

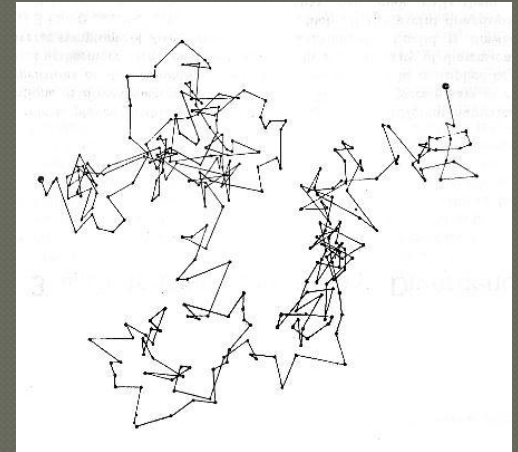
Impaction

● Definition

- occurs when air containing particles moves past stationary objects and some of the airborne particles collide with objects and stick

● Examples

- Dirt accumulation on fan blade leading edges
- Crushed insects on windshields
- Accumulation of snow on upwind side of a tree during a snowstorm



● Small particles

- May stick to solid surfaces after diffusing through a stagnant air boundary layer above the solid (Brownian motion)
- Deposition velocity model – typically in the range of 0.1 to 1 cm/sec for many atmospheric particles

Impaction

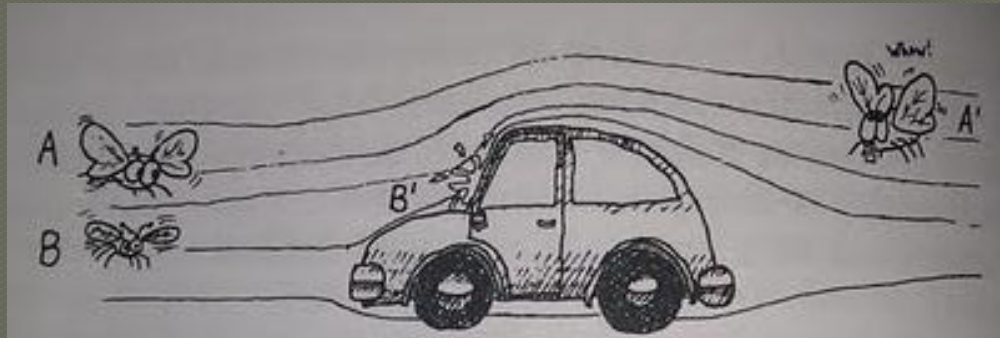
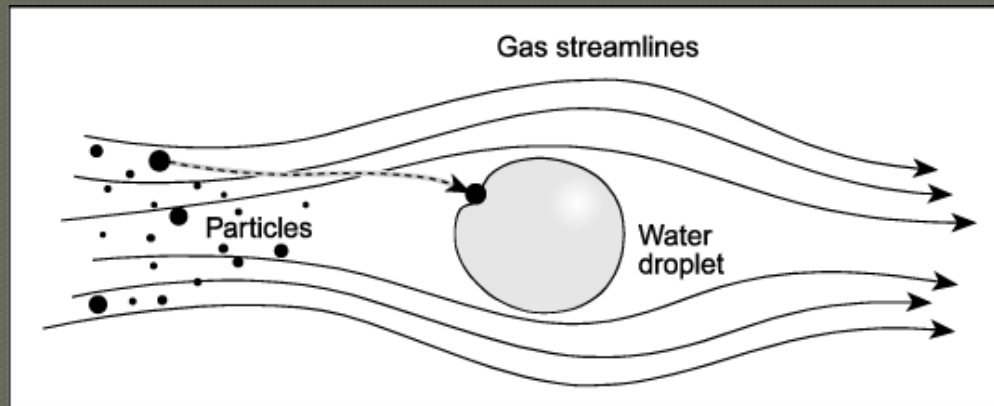


Figure 4-30



Source: U.S. Environmental Protection Agency in collaboration with North Carolina State University at

http://en.wikipedia.org/wiki/File:Impaction_scrub.gif

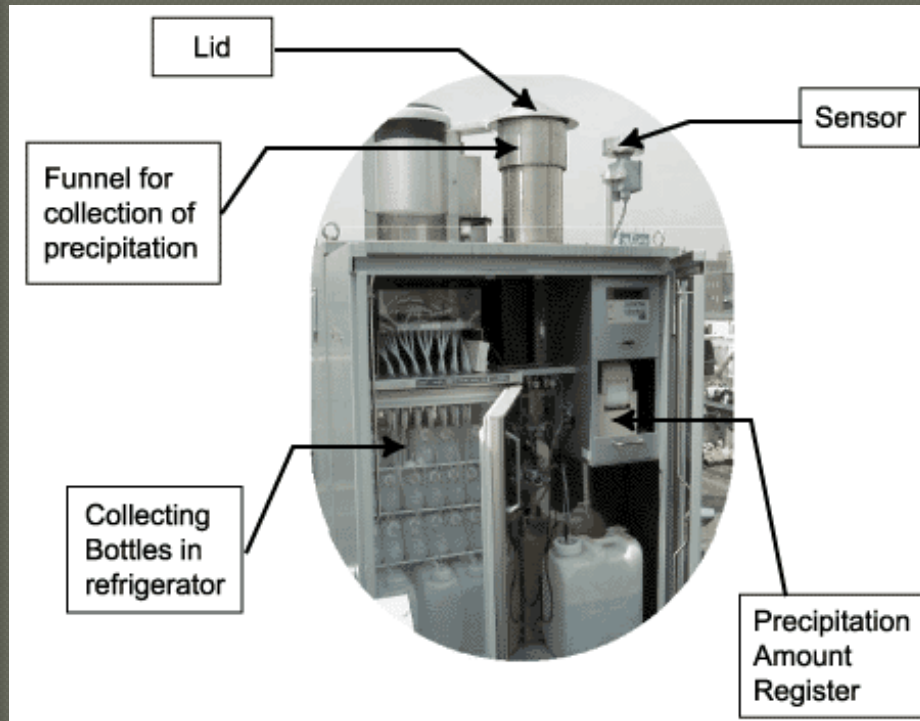
Absorption of Gases

- Many atmospheric gases are absorbed, by liquid or solid surfaces
- Usually described by a thin film model (figure 4-31)
- Flux of vapor into the surface:
 - $J = (C_a - C_s) * D / \partial$ where:
 - J = flux density of vapor [M/L^2T]
 - C_a = [vapor] in air at solid surface [M/L^3]
 - D = diffusion coefficient in air [L^2/T]
 - ∂ = thickness of the stagnant air boundary layer [L]
- Deposition velocity (V_d) = piston velocity describing the deposition of a chemical from the air into a liquid or solid surface by absorption

Deposition Velocity (V_d)

- Commonly estimated by an empirical modeling approach (inferential modeling)
 - Uses meteorological data and information on the surface characteristics of the vegetation (figure 4-32)
- Deposition velocities for absorption often found to be similar in magnitude to values for impaction
 - ~0.1 to 1cm/sec for gases whose absorption into a surface is limited by the rate of Fickian transport through a stagnant boundary layer

Wet Deposition



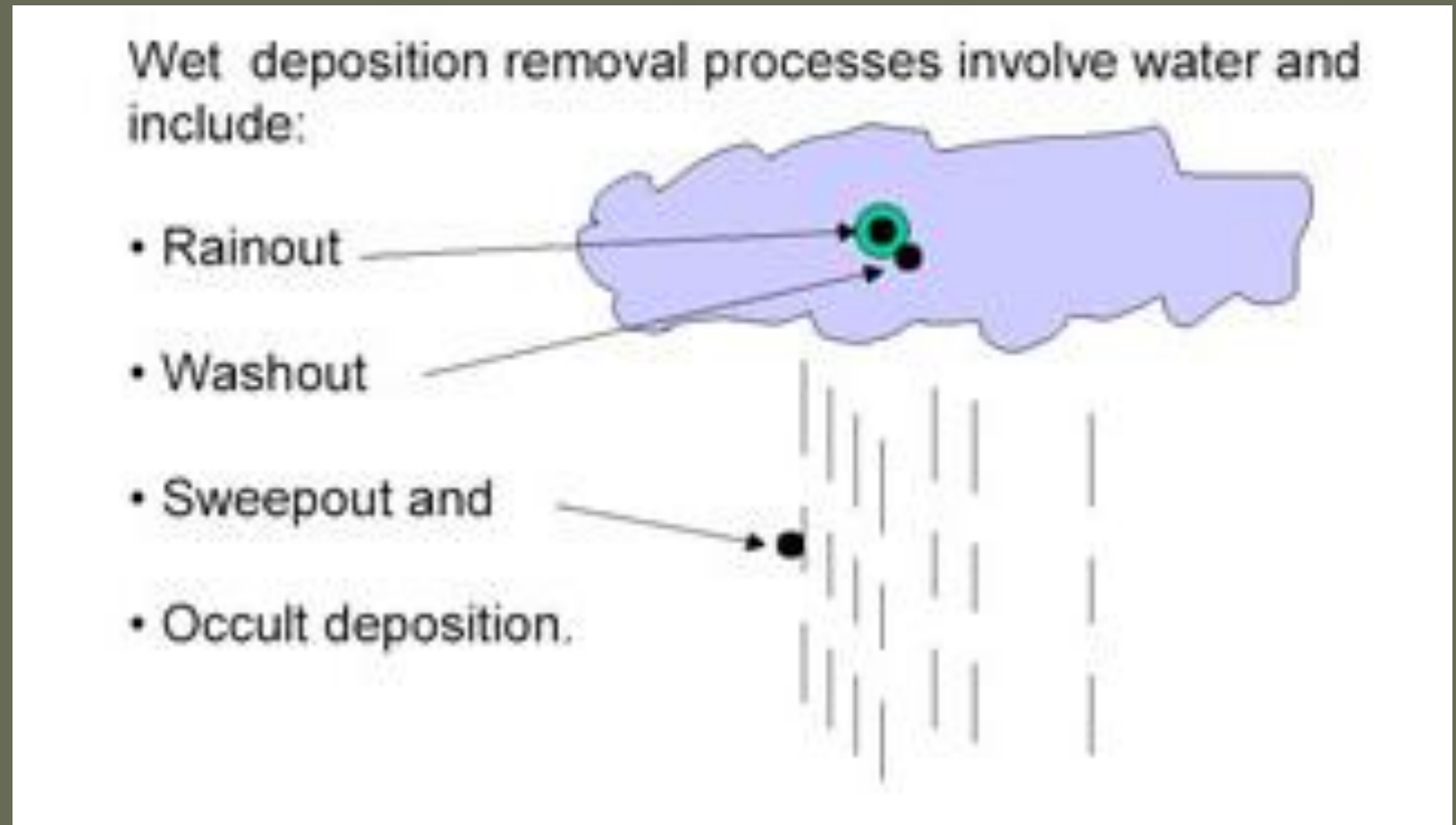
Source:

http://www.menlh.go.id/apec_vc/osaka/eastjava/acid_dep_en/sampling.html

- Atmospheric chemicals are accumulated in rain, snow or fog droplets and deposited on Earth's surface
- Removes chemicals whose rates of settling, impaction or absorption are slow to zero
- **Rainout:** incorporation of chemicals into water droplets in a cloud
- **Washout:** incorporation occurs beneath a cloud and ppt falls towards Earth

Rainout: Particles act as nuclei for cloud condensation
Washout: Removal of particles by cloud droplets
(difference consists in the pre-existence of a collecting drop)

Source: http://www.dustsuppression.it/dust_suppression_works.php?pag=6



Wet Deposition of Gases & Vapors

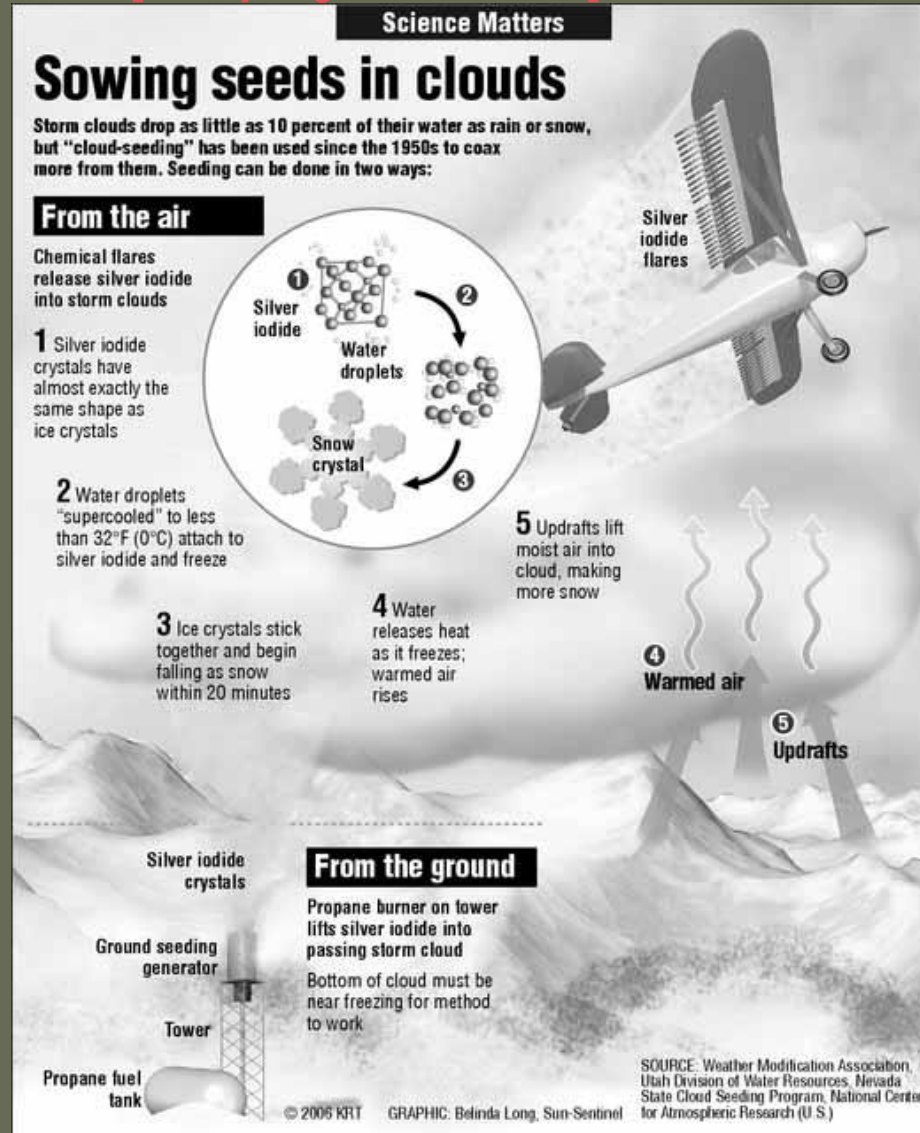
- Gases and vapors can be removed by dissolving in raindrops
- [chemical], at equilibrium, is given by Henry's Law
 - $C_{\text{water}} = C_{\text{air}}/H$ where
 - $C_{\text{water}} = [\text{chemical in water}] \text{ (M/L}^3\text{)}$
 - $C_{\text{air}} = [\text{chemical in air}] \text{ (M/L}^3\text{)}$
 - H = Henry's law constant
- Chemical equilibrium may be assumed:
 - Rain forms contact with air (rainout)
 - Travels through several tens of meters in contact w/air mass (washout)
- $C_{\text{water}}/C_{\text{air}}$ is the washout ratio [Table 4-10]

Wet Deposition of Particles

- Major mechanism by which particles are incorporated into ppt is by serving as nucleation sites for condensation at the onset of water droplet or ice crystal formation
- Can also be incorporated into already-formed water droplets w/in a cloud by collision
- Removal of particles by rainout is far more effective than the dry deposition of particles

Wet Deposition of Particles

Source: <http://coyotegulch.files.wordpress.com/2010/03/cloudseedingexplained.jpg>



Atmospheric Chemical Reactions

- Most driving directly or indirectly by sunlight
- Photochemistry more important in atmosphere vs. surface waters bc:
 - Higher levels of illumination
 - Minor competing transformation processes
 - Significant flux of radiation at the energetic shorter wavelengths below 290nm
- Water droplets and PM influence the rates of transformations in the atmosphere
 - Homogeneous rxn: only gaseous chemicals
 - Heterogeneous rxn: gas in conjunction with a liquid phase or solid surface

Oxidants in the Troposphere

- **Ozone** (one of best known atmospheric oxidants)
 - At ground level: associated with lung irritation and plant damage, degrades materials (i.e. rubber windshield wiper blades) and has deleterious impact on agriculture
 - Stratosphere: blocks UV rays
- **Free radicals**
 - Species that contain an unpaired electron
 - e.g. hydroxyl, hydroperoxy, alkoxy and alkyl peroxy

Photochemical Smog



Harris County, Houston, TX.

Source: <http://www.hcphe.org/eph/ozone/home.htm>

Dr. Haagen-Smit



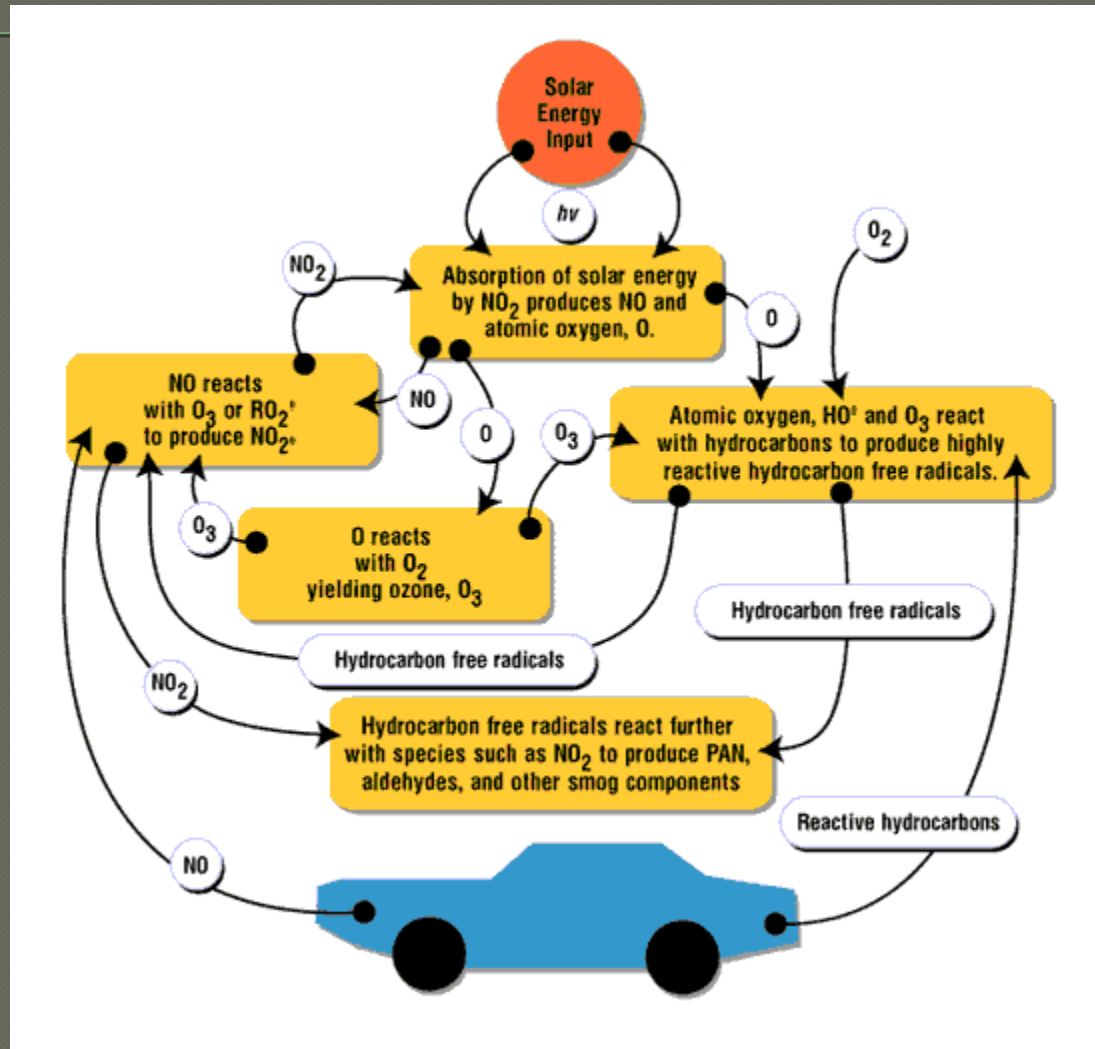
Arie Jan Haagen-Smit

1900–1977

- “analyzer of pineapple volatiles”
- Credited with the discovery of the chemical nature of LA’s smog (1952)
 - Used rubber strips to test for cracking
- First chairman of the CA Air Resources Board

Source: <http://www.arb.ca.gov/research/hsawards/japcaarticle1978.pdf>

Photochemical Smog



PAN =
peroxyacetyl
nitrate

$\text{O}_3\text{-NO}_x$ Cycle

Species involved:

- Nitrogen dioxide (NO_2)
 - only species that absorbs light
 - sometimes visible as a brownish haze
- Nitric oxide (NO)
 - automobile & industrial origin
- Molecular oxygen (O_2)
- Ozone (O_3)

Key reactions:

- $\text{NO}_2 \xrightarrow{\text{light}} \text{NO} + \text{O}$ [4-33]
- $\text{O} + \text{O}_2 \rightarrow \text{O}_3$ [4-34]
- $\text{O}_3 + \text{NO} \rightarrow \text{NO}_2 + \text{O}_2$ (dark) [4-35]
- 1 molecule of NO_2 consumed for every molecule O_3 produced so the maximum amount of O_3 can never be greater than the initial amount of NO_2

....but

Measured $[O_3] > [O_3]$ predicted by the O_3 - NO_x cycle

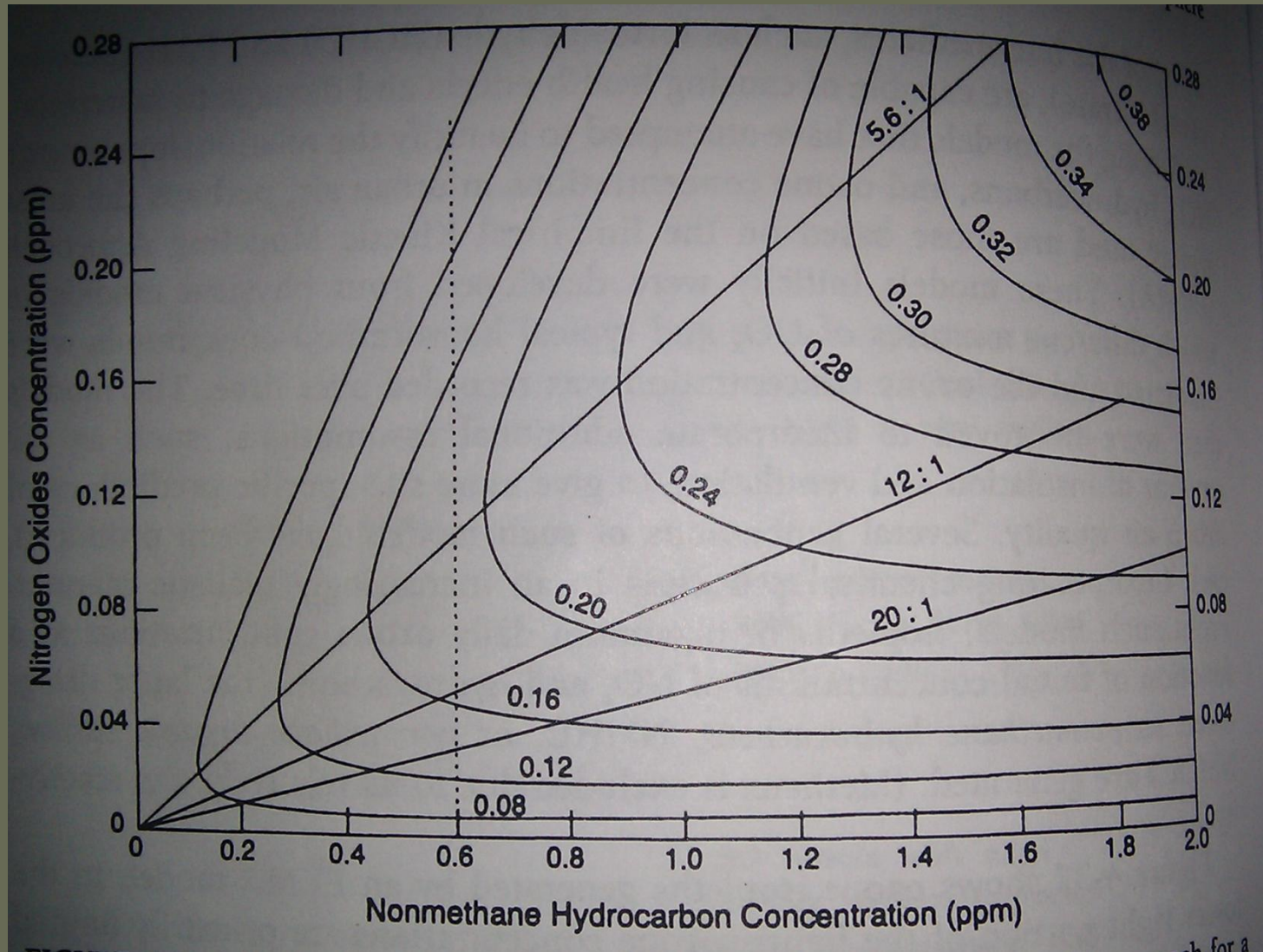
● Why?

- Other photochemical oxidants (formed from hydrocarbons) reoxidize NO to NO_2 w/out consuming ozone (e.g. Figure 4-36)

● Models attempt to quantify the relationships among NO_x , hydrocarbons, and O_3 in urban air

- Empirical Kinetic Modeling Approach (EKMA)
- Figure 4-37 (next slide)

Figure 4-37



Acid Deposition

- Acid deposition

- Deposition of acidic substances from the atmosphere onto the Earth's surfaces
- Widespread attention throughout the 1980s
- Caused acidification of some lakes and streams, and corresponding loss of fish populations

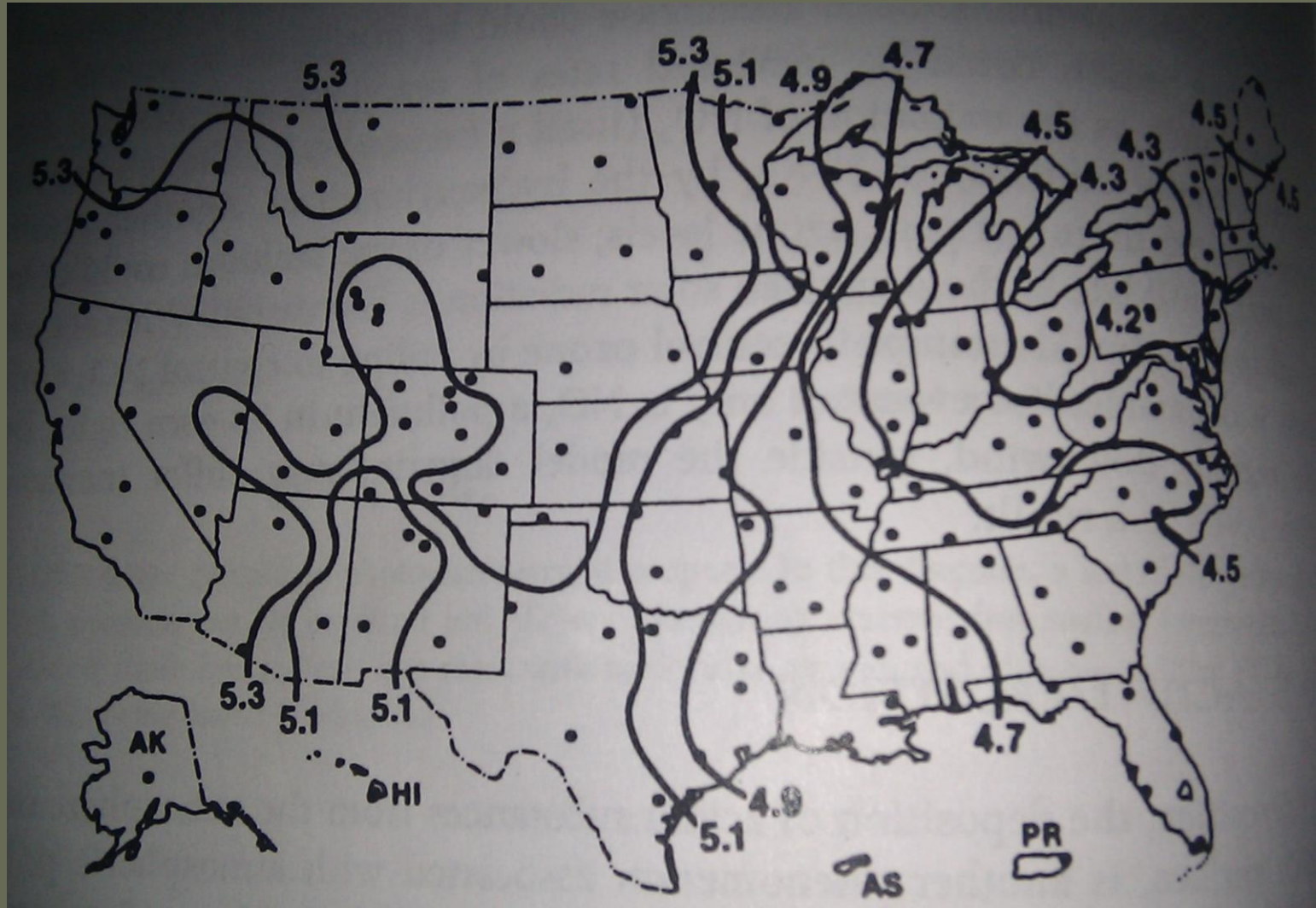
- Acid rain

- Low PH of precipitation that has been observed across the US (figure 4-38)

- Very costly to decrease emissions of acid-producing pollutants

Figure 4-38. Mean pH of ppt of the US in 1990 (based on weekly measurements at 170 sites)

NB: “unpolluted rain pH” ~ 5.7





Many trees in the Great Smoky Mountains, such as those in the foreground, have succumbed to the effects of acid rain.

Source: *Encyclopædia Britannica Online*. Web. 24 Mar. 2011..

<http://www.britannica.com/EBchecked/media/106953/Many-trees-in-the-Great-Smoky-Mountains-such-as-those>

History of Acid Rain in the US

- 1980 – US Congress passes Acid Deposition Act (10-yr research program)
- 1991 – first assessment of US acid rain
- 1990 – amendments to Clean Air Act passed (limit SO₂ emissions from power plants)
- 2005 – Clean Air Interstate Rule (states with a solution to power plant pollution drift)
- Since 1990s, SO₂ emissions dropped 40%



Impact of Acid Rain on Statues

Source: http://en.wikipedia.org/wiki/File:Pollution_-_Damaged_by_acid_rain.jpg

Acid Fogs

- pH of water reaches 3 or lower (H_2SO_4 can be surprisingly high)
- perhaps one reason forest damage often more pronounced at higher elevations

Source:

http://www.swissinfo.ch/eng/Home/Archive/Acid_fog_found_to_pollute_forests.html?cid=2981522



Innocuous looking fog can actually be more harmful to forests than acid rain (Keystone)

Acid Sources

- 2 largest contributors
 - Sulfuric acid (H_2SO_4)
 - Nitric acid (HNO_3)
- Most not emitted directly into the atmosphere, instead emissions contain acid precursors
 - SO_x : sulfur-containing impurities in fuel (coal and residential fuel oils)
 - NO_x : nitrogen-containing impurities in fuel and rxns btwn atmospheric nitrogen and oxygen at elevated temps in fuel-burning equipmt (industrial boilers, stationary power plants, and automobile engines)
 - Both are irritating air pollutants and fine particle precursors
- Sulfur oxidation
 - Bulk occurs by heterogeneous processes in water droplets ($\sim 0.3/\text{hr}$)
 - Rainout and washout processes contribute to sulfuric acid (H_2SO_4)

Acid Sources, cont.

● Nitric Acid Deposition

- Similar to sulfuric acid deposition
 - i.e. emission of acid precursors that are subsequently oxidized in the atmosphere to form strong acids
 - Both rainout and washout processes contribute to the incorporation of HNO_3 into ppt
-
- Accurately estimating the total amount of acid deposited on a receptor (i.e. lake or forest) is problematic bc acids are deposited by difficult to quantify dry and wet processes

Environmental Fate of Acid Deposition

- Two key questions:

- What happens to the acidic chemicals deposited on a watershed?
- To what extent does acid deposition change the pH of surface waters?

- Multiple answers to these questions!

- Behavior of nitric acid is more complex than that of sulfuric acid bc nitric acid is frequently a limiting plant nutrient

Stratospheric Ozone Chemistry

- O_3 production:
 - $\text{O}_2 + h\nu \rightarrow 2\text{O}$
 - $\text{O} + \text{O}_2 \rightarrow \text{O}_3$
- O_3 can be destroyed by interaction with another photon that breaks it into O_2 and O or w/ rxn w/ other species like NO and chlorine atoms.
- Produced at maximum rates in equatorial regions where solar radiation most intense
- O_3 occurs as a broadly distributed gas whose [peak] occurs midstratosphere
- Entire O_3 shield equivalent to a layer of O_3 2-4mm thick at sea level pressure

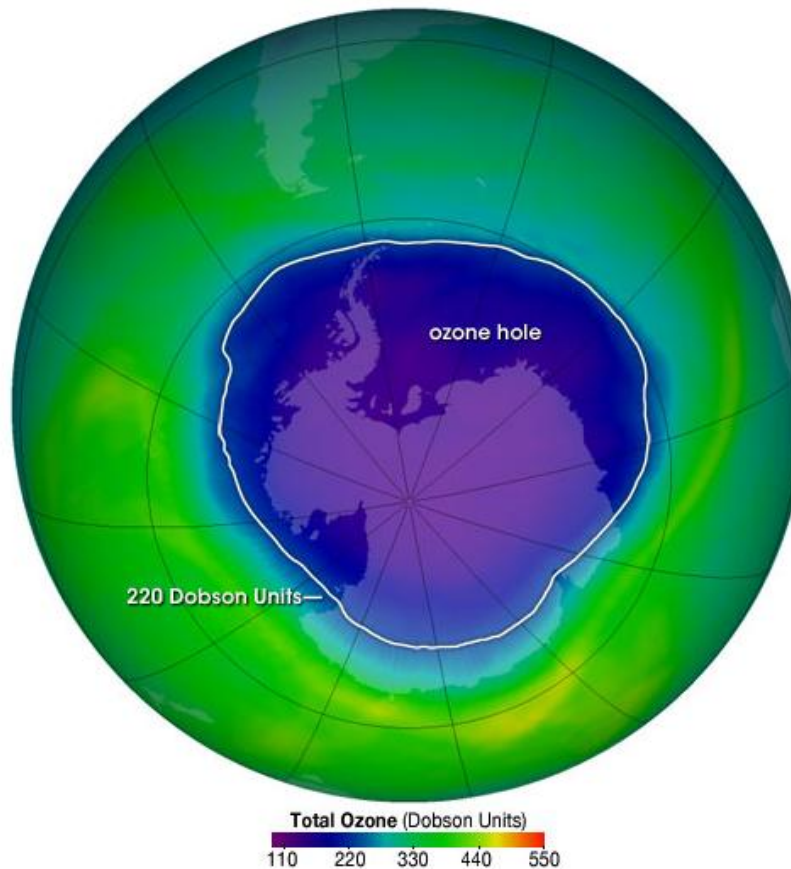
Ozone Hole

- Dobson unit (most common unit for measuring $[O_3]$)
 - 1 Dobson unit = # molecules O_3 to create a layer of pure O_3 0.01 mm thick at $0^\circ C$ and 1 atm
- Discovery of O_3 hole
 - Team from British Antarctic Survey, using ground-based instrumentation, reported that mean $[O_3]$ decreasing
 - American investigators reanalyzed satellite data – data erroneously discarded!
 - Now acknowledged to be result of the catalytic degradation of O_3 by chlorine (source CFCs)
- Montreal Protocol (1987)
 - "perhaps the single most successful international agreement to date has been the Montreal Protocol"
– Kofi Annan
 - Ratified by 196 countries

Ozone Hole

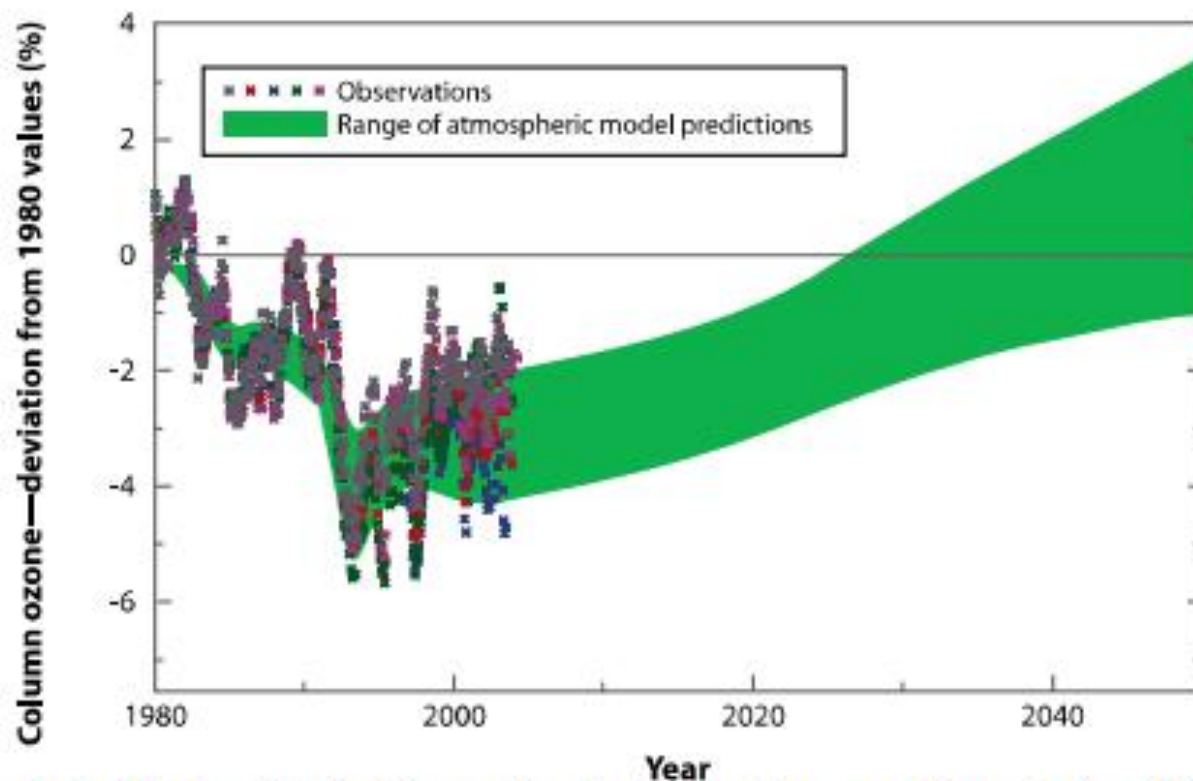
Source:

<http://ozonewatch.gsfc.nasa.gov/facts/hole.html>



The ozone hole is the region over Antarctica with total ozone of 220 Dobson Units or lower. This map shows the ozone hole on October 4, 2004. The data were acquired by the [Ozone Monitoring Instrument](#) on NASA's [Aura](#) satellite.

Global Ozone Depletion and Recovery



Source: EPA: *Achievements in Stratospheric Ozone Protection - Progress Report*. April 2007.

Source: http://www.epa.gov/ozone/downloads/MP20_Background.pdf

Homework

- #31
- #15